

# Independent Research

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*AI evaluation · Formal systems · Reproducible mathematics*

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## Research focus and contribution

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Research on preserving evidence, dependencies and correction state across long-running AI-assisted work. Current outputs include a finite correction-state model, evaluation-development infrastructure, three public mathematical preprints and an exact computational certificate.

**Research direction:** problem selection, task framing, verification requirements, scrutiny of unsupported claims, correction decisions and release scope.

**AI assistance:** substantial contributions to exploration, formalization, proof and code development, computation, checking, literature synthesis and writing.

## Formal systems and AI evaluation

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### Correction-sufficient state

Epistemic Transition Integrity (ETI) · Finite technical note and reference code · Release under review

A finite interval-edit model characterizes when one correction is valid across all states compatible with retained information. A related conflict-graph formulation characterizes the minimum one-shot code count. Grounded alternative support preserves conclusions when one justification is withdrawn but another survives.

**Evidence:** the reference suite covers 32,768 support states and 98,304 transitions, including 49,152 removal scenarios, within the specified twelve-rule template family. These are finite implementation checks, not model-performance trials.

[Model, assumptions and research status](#)

### Premise-Governed Context Scheduling

Study 14 · Evaluation design and development instruments · No comparative model outcome yet

The four-condition design compares full context, status-labeled full context, semantic retrieval and premise-governed scheduling. Independent operational-state construction, matched source information, assessment independence and full construction/maintenance costs are explicit requirements.

**Development result:** a successor context-parity checker rejects changed source text that the earlier identifier-and-order check accepted. This establishes a specific checker improvement, not an empirical reliability gain.

[Evaluation design and status](#)

## Verification and reproducibility

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Methods include exact arithmetic, proof reconstruction, implementation-distinct checks, adverse fixtures, versioned corrections and source-bound releases. The evidence record separates mathematical proofs, finite computations, software tests and empirical outcomes.

**Verification status:** checks are internal and method-diverse; they do not constitute external specialist peer review. General ETI efficacy and consistent online maintenance of compressed correction state remain research questions.

[Evidence and corrections](#) · [Methods and AI-use disclosure](#)

# Public preprints and certificate

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*Three theorem manuscripts and one pointwise computational result*

The linked releases contain arguments, code or certificates and verification records. Journal publication, arXiv posting and completed external specialist review are not recorded in the current public status.

## Explicit Counterexample Families to Five Finite-Group Inequalities from Graffiti<sup>3</sup>

Finite-group theory · Public preprint · 2026 · v1.0.0-preprint

Unitriangular and dihedral families refute C20–C22 and C27–C28 of the specified Graffiti<sup>3</sup> source version. Symbolic family arguments carry the general claims; exact arithmetic, certificate checks and mutation controls supply computational corroboration. Priority and significance remain subject to specialist assessment.

[Manuscript](#) · [Frozen release and replay](#)

## Largest-Index Zonotope Bounds for Planar Projections of Semiorder Ideals

Combinatorics and geometry · Public preprint · 2026 · v1.0.0-preprint

A largest-index decomposition gives all-size projection bounds for represented semiorders under explicit geometric conventions. The exact finite verifier covers 318,667 cases; the proof, rather than enumeration, supports the general statement. Unrestricted parametric closure remains open.

[Manuscript](#) · [Frozen release and replay](#)

## Excess-Degree Bounds for Minimal Counterexamples to the Erdős–Gyárfás Conjecture

Graph theory · Public preprint · 2026 · v1.0.0-preprint

For a counterexample chosen first by minimum vertex count and then by minimum edge count, the manuscript derives  $|A| \geq 2b + s + 4$  under its degree-class definitions. The structural proof is separate from other bounded computational packets; the global conjecture remains open.

[Manuscript](#) · [Frozen release and replay](#)

## A 104-transition totient trajectory

Number theory · Public computational certificate · 2026 · v3.0.0-f104

The release establishes  $F(400000287233629) = 104$  for iteration of  $\varphi(n) + 1$  to the first prime. The trajectory is paired with evaluators, factor-from-scratch checks and mutation tests. This is a pointwise result, not a maximum, priority or unboundedness claim.

[Certificate and replay](#)

## Education and research environment

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California State University, Northridge · May 2025

B.S. Geography and Environmental Studies · Minor in Geology · Cum Laude

**Tools used in AI-assisted projects:** Python, Git/GitHub, LaTeX, SymPy and GAP/SmallGrp. Tools are listed as part of the project environment.

**Languages:** English and Norwegian (fluent); strong comprehension of Swedish and Danish.